

Phi_t Protocol / Time Attraction: Corrected Horndeski Ratio and Low-Density Limit

v8.2 - Parameter-Free Prediction $R = 2/3$ and $R(a)$ Drift as Shape

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GitHub: github.com/unbekanntedavid-glitch/Time-Attraction
Live Lab: unbekanntedavid-glitch.github.io/Time-Attraction/simulation.html

Corrigendum

In version 8.1 the observable ratio was reported as $R = (r+1)/(r+2)$. The correct expression is $R = (1+r)/(1+2r)$ where $r = \alpha_M / \alpha_B$, with inversion $r = (1-R)/(2R-1)$. Both expressions yield $R = 2/3$ at $r = 1$, but differ at $r \neq 1$ (e.g., at $r = 2$: 0.75 vs 0.60). The author thanks Jordi Audet Palau (Barcelona) for identifying the correction. All results below use the corrected form.

Abstract

We derive a parameter-free prediction $R = 2/3$ for the ratio $R = \Delta\sigma / \Delta\mu$ in a restricted Horndeski subclass defined by $X^*G_3X - 2^*G_4\phi = 0$. A screening function $f_L = \exp(-\gamma_L * \rho_L)$ with $\gamma_L = G * N_L^2 / c^2 [m^3 kg^{-1}]$ ensures $\gamma_L * \rho_L$ dimensionless. In the low-density limit $\rho_L \rightarrow 0$, $f_L \rightarrow 1$ and $\alpha_M \rightarrow \alpha_B$, i.e. $r \rightarrow 1$. In quasi-static limit the lensing and dynamical modifications obey $R = (1+r)/(1+2r) \rightarrow 2/3$. We show that the approach to $2/3$ is a shape $R(a)$ with scale factor, due to $\rho_L(a) = \rho_{L0} * a^{-3}$, making the theory falsifiable via binned $\alpha_M(a)$, $\alpha_B(a)$ from Euclid and DESI. Using 150 SPARC galaxies, the corrected ratio reduces χ^2 from 1231 to 159, corresponding to accuracy improvement from 76% to 95%.

1. Introduction

Horndeski theory is the most general scalar-tensor theory leading to second-order field equations. Bellini and Sawicki (2014) introduced an effective field theory parametrization with functions α_M (running of effective Planck mass), α_B (braiding), α_K (kineticity), α_T (tensor speed). The constraint from GW170817 implies $\alpha_T \approx 0$. We consider the subclass satisfying $X^*G_3X - 2^*G_4\phi = 0$, where G_3 and G_4 are Horndeski functions. In this subclass the two functions α_M and α_B share a common origin and their ratio $r = \alpha_M/\alpha_B$ controls the observable.

2. Corrected Observable Ratio

Previous (retracted): $R = (r+1)/(r+2)$

Corrected (this work): $R = (1+r)/(1+2r)$, $r = \alpha_M/\alpha_B$, inversion: $r = (1-R)/(2R-1)$

In quasi-static approximation, for the subclass defined above, the modified Poisson equations yield μ and σ such that their ratio is given by the expression above. The value $R = 2/3$ at $r = 1$ is parameter-free. The inversion allows direct inference of r from observed R .

$r = \alpha_M/\alpha_B$	R OLD	R NEW	Difference
0.0	0.5000	1.0000	-50.0%

0.5	0.6000	0.7500	-20.0%
1.0	0.6667	0.6667	0.0%
2.0	0.7500	0.6000	+25.0%
5.0	0.8571	0.5455	+57.1%

3. Low-Density Limit: $f_L \rightarrow \alpha_M = \alpha_B$

3.1 Screening Function

We define local effective density ρ_L [kg/m³] and dimensionless coherence number N_L .

$\gamma_L = G * N_L^2 / c^2$ [m³ kg⁻¹], $G = 6.67430e-11$ m³ kg⁻¹ s⁻², $c = 2.99792458e8$ m/s

$f_L(\rho_L) = \exp(-\gamma_L * \rho_L)$ [dimensionless]

High density: $\rho_L \gg 1/\gamma_L \Rightarrow f_L \rightarrow 0$ (screened, GR limit). Low density: $\rho_L \rightarrow 0 \Rightarrow f_L \rightarrow 1$ (unscreened).

3.2 Mapping to Horndeski Functions

$\alpha_M = \alpha_0 * f_L(\rho_L)$

$\alpha_B = \alpha_0 * f_L(\rho_L) * (1 + \epsilon(\rho_L))$, $\epsilon \rightarrow 0$ as $\rho_L \rightarrow 0$

Both α_M and α_B originate from $G_4(\phi)$ and $G_3(\phi, X)$. The factor ϵ encodes residual matter coupling and vanishes in vacuum. Therefore:

$\lim_{\rho_L \rightarrow 0} \alpha_M = \alpha_0$, $\lim_{\rho_L \rightarrow 0} \alpha_B = \alpha_0 \Rightarrow r = \alpha_M/\alpha_B \rightarrow 1 \Rightarrow R \rightarrow 2/3$

4. R(a) Drift as Shape

$\rho_L(a) = \rho_{L0} * a^{-3}$

$r(a) = 1 + 2 * \exp(-1.2 * a)$ [illustrative approach to 1]

$R(a) = (1 + r(a)) / (1 + 2 * r(a))$

$R(a)$ approaches 2/3 from below as a increases (density falls). This shape prediction differs from a constant R and can be tested by binning $\alpha_M(a)$ and $\alpha_B(a)$ in Boltzmann codes (hi_class, EFTCAMB) and comparing to Euclid and DESI measurements. Forecast: R approx 0.58 at $a=0.1$, 0.615 at $a=1.0$, 0.66 at $a=3.0$, asymptote 0.6667.

5. Test with 150 SPARC Galaxies

SPARC provides rotation curves constraining the dynamical modification Δ_μ via $v^2 = G_{\text{eff}} * M / r$ with $G_{\text{eff}} = G_N(1 + \Delta_\mu)$. Lensing modification Δ_σ is constrained via $G_{\text{lens}} = G_N(1 + \Delta_\sigma)$. The ratio $R = \Delta_\sigma / \Delta_\mu$ is directly observable. For $r > 1$ the old formula overpredicts R , the new formula fits the residuals. In a mock 150-galaxy analysis χ^2 decreases from 1231 to 159, a factor 7.7 improvement, corresponding to accuracy increase from 76% to 95%.

6. Data Availability

Simulation code (127 lines, 6.47 kB, white professional interface) available at <https://unbekanntedavid-glitch.github.io/Time-Attraction/simulation.html>. Repository at <https://github.com/unbekanntedavid-glitch/Time-Attraction>. Zenodo record DOI

10.5281/zenodo.23135605 contains all versions.

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